Molecular Reality April 22, 2026

Motivating Questions

- The shift in the standing of atomic-molecular theory ca. 1910 is often treated as a paradigm of successful scientific reasoning about unobservables. What does this case actually reveal about the nature, scope, and limits of scientific knowledge?
- *Smith and Seth vs. van Fraassen*: existing philosophical assessments fail to adequately characterize the evidential reasoning in this case. Doing so requires a more fine-grained account of *theory-mediated measurement* (Smith and Seth), and the resulting account diverges from van Fraassen's empiricist counter-narrative, regarding what is relevant in such an account and what it establishes.
- How does this case bear on broader themes we have been exploring: the relation of theory to measurement, the structure of evidence, and whether successful measurement practices can underwrite a form of scientific realism?

Background: Brownian Motion and Perrin's Experiments

- *Atomic / molecular hypothesis*: liquids and gases are composed of an enormous number of localized, interacting parts. Different aspects: chemical composition, kinetic theory (heat as motion of constituents), radiation, etc. Longstanding criticisms even among those who granted success of the kinetic theory of *gases* (as opposed to fluids, solids, ...).
- *Brownian motion*: irregular, unceasing, complex motion of granules in dilute emulsion or colloidal suspension (first observed by Robert Brown, 1827). Perrin undertook high-precision observations of three aspects of the motion of granules of varying size (using the ultramicroscope, invented 1902): (1) vertical gradient (due to gravity), (2) translational motion, (3) rotational motion.

  Einstein and Smoluchowski had independently developed a theory of Brownian motion that relates properties of the motion (e.g., mean displacement) to N_0 and other measurable parameters; Langevin (1908) gives an alternative derivation without assuming molecular-kinetic theory.
- Perrin's experiments establish that the mean kinetic energy of the granules varies linearly with absolute temperature, $E = W_T T$, with W_T a *universal* constant (independent of fluid, size of granules, etc., within the range tested). They also establish subtler properties, e.g., that displacements follow a Gaussian distribution.
- *Two inferences* (following Smith and Seth):
  1. Observable quantities (vertical, translational, rotational displacements) $\rightarrow$ quantities of the *granules* (e.g., mean translational kinetic energy). Based on Einstein's "random walk" theory (or Langevin 1908).
  2. Quantities characterizing granules $\rightarrow$ quantities of the *molecules of the fluid* (including N_0). The "weak link": *assume* that the mean translational kinetic energy of the granules equals that of the molecules.
- *Other measurements of* N_0: at around the same time as Perrin's Brownian motion work, there were a number of new methods for determining N_0 at high precision (from ions, α-particles, black-body radiation) as well as more qualitative estimates. In the closing passage of *Atoms*, Perrin emphasizes agreement of values from 16 different empirical sources.

van Fraassen, "The Perils of Perrin"

Counter-narrative against the realist "lore" surrounding Perrin: minimal, empiricist account of the development of kinetic theory as motivated by efforts to provide *empirical grounding*, opposed to a realist reading ("strange reading") that focuses on vindicating existence claims.

- *Historiographical critique*: (i) the tendency to recast debates in contemporary terms; (ii) mischaracterization of positions (energeticists and critics of kinetic theory *also* employed "unobservable entities"); (iii) the degree to which the atomic hypothesis was genuinely an open question by Perrin's time.
- *Foundational problems with kinetic theory*: *not* primarily skepticism about unobservables, but rather the inability to assign consistent values to fundamental parameters (atomic weights, specific heats, degrees of freedom).
- *Empirical Grounding* (adapted from Weyl):
  1. *Determinability*: any theoretically significant parameter must be such that there are conditions under which its value can be determined on the basis of measurement.
  2. *Concordance*: (i) *Theory-Relativity* — this determination can, may, and generally must be made on the basis of theoretically posited connections; (ii) *Uniqueness* — the quantities must be "uniquely coordinated": there must be concordance in the values determined by different means.
  3. *Non-Triviality*: there must be alternative possible outcomes for the same measurements that would serve as evidence *against* the theory, based on the same theoretically posited connections.
- *Theory-dependence of grounding* and the continual elaboration / extension of theory to make grounding possible. Van Fraassen assesses Perrin's contribution in terms of the number of unknown parameters needed in the description of molecules in a gas, and the new relationships required to enable their determination from empirical results. On his account, the historical core of molecular-kinetic theory was not "informative enough to make testing possible at all"; it had to be progressively *enriched* so as to allow experiments that could determine the values of its parameters. Once grounding is achieved, it is the *whole theory* that becomes testable.
- Perrin's closing passage — rewriting relationships to eliminate theoretical quantities in favor of measurable ones — read as the culmination of this grounding effort, *not* as support for the "reality" of molecules in any stronger sense than empirical adequacy of models.

Smith and Seth, *Brownian Motion and Molecular Reality*

A more fine-grained reconstruction of Perrin's reasoning, similar in some ways to Smith's study of Newton but in a quite different setting:

- *Longue durée* evidential reasoning: evidence becomes available at different stages of long-term research programs, different kinds of considerations are relevant at different stages, an iterative conception of inquiry is required, and detailed historical case studies are indispensable.
- *Gaining reliable access to novel quantities*:

  - "Taking a hypothesis to be true": support new lines of research, typically introducing novel unobservable quantities that enable the design of experiments, the formulation of regularities, and the development of mathematical theories.
  - Different kinds of risk: (i) reifying quantities that are artifacts or fictions; (ii) accepting a hypothesis that is insulated from further critical evaluation (a "garden path"); (iii) relevant quantities remain beyond experimental reach; (iv) obtaining clear results requires assumptions that conflict with other aspects of physics.

- *Fine-grained assessment of evidential reasoning*:
  - Contrast with the monolithic "Theory T implies E" approach: local assessment of specific elements of theory needed for particular arguments, with differential support for different components.
  - Detailed evaluation of what specific lines of evidence establish, which aspects of theory they actually depend on, and possible limitations or sources of error.

Successful Measurement Practice (SS Ch. 4 and Postscript §P.3)

Levels of representation (developed using terminology from van Fraassen's *Scientific Representation*):

1. *Data*: values obtained from individual observations, generally expressed in terms not accessible to direct observation (e.g., the magnitude of an electric current, viscosity, or the radii of Perrin's granules, which were inferred indirectly rather than placed against a scale).

2. *Data Models* (Suppes): "distillation" of data through statistical techniques, including accounting for systematic errors. (E.g., extracting a single radius value for the granules from multiple indirect measurements; squaring and averaging displacements to obtain ξ^2.)

3. *Measure*: an equation relating a number of distinct physical quantities — typically a target quantity (such as W or N_0) alongside other more accessible quantities. For Perrin's translational displacements: $\xi^2\tau = Wa/9\pi\zeta$.

4. *Mathematical Theory*: equation typically derived from some underlying physical theory (e.g., Einstein's treatment of Brownian motion).

5. *Explanatory Theory*: "vision of reality we use to explain why the equations work" (Weinberg).

At what level of representation can we feasibly establish permanence? How does this position regarding permanence relate to other realist positions (e.g., entity realism, structural realism)?

Three demands on a successful measurement practice:

- *Stability*: the equation (the measure) holds for a wide range of values of variable, manipulable quantities. The assessment of stability is *independent* of the underlying theory: experiments establish a regularity, although the *interpretation* of the quantity depends on theory. (For Perrin: experiments varying radius of granules, viscosity, etc. yield constant W; theory is required to interpret this as mean kinetic energy of the granules.) Stability provides "the first layer of evidence in support of the premises entering into [the measure]"; when a measure is not stable, those premises are subject to question.

- *Convergence*: agreement of values for a target quantity appearing in distinct stable measures. (For Perrin: *three* distinct stable measures for W — vertical gradients, translational displacements, diffusion coefficients — whose only link is the theoretical derivation; convergence supports not just the measure but whatever aspect of the theoretical derivation establishes the link.) Convergence thereby provides *stronger evidence for the reliability of the differing constitutive premises than stability alone.*

- *Amenability to increasing precision*: limits to precision stem from limited data or data models rather than from imprecision in the derivation of the measure or in our theoretical understanding of the quantity. (Supports identification of systematic errors in converging measures — e.g., the 10% error in Perrin's Brownian motion experiments.)

Permanence: response to the anti-realist skeptical challenge — "whether any evidence for propositions in science can ever eliminate virtually all possibility of future information refuting them" (Postscript, p. 392). Response:

> *[T]he demands for stability of a measure, its convergence with other measures of nominally the same quantity, and its amenability to increasing precision* ***can*** *eliminate virtually all possibility of future information showing that the* ***value*** *itself obtained from it was all along just an artifact of the data or data model from which the value was extracted, and therefore all along not something corresponding, at least to a degree of approximation, to a bona fide physical quantity.* (emphasis in original, Postscript, p. 405)

Note the strong emphasis on *value* rather than physical interpretation. The "can" acknowledges that this does not happen automatically but depends on the strength of the evidence for the three demands and on specifics of the measure.

Perrin Revisited: The Four Primary Quantities and Their Weak Links

A central claim of Smith and Seth is that Perrin's case is *not* appropriately described as showing that molecular-kinetic theory as a *whole* is tested by his experiments. Rather, the evidence bears differentially on different components of the theory. Their diagnostic tool is the distinction between *primary quantity* and *linking premise.*

Common logical structure: Each determination of N_0 in the period 1908–1912 involved:

- A theory-mediated measurement of some *primary quantity*, whose measure could be assessed for stability (and sometimes convergence) *without any constitutive commitment to molecular-kinetic theory.*

- A *further "linking" premise* that connected the primary quantity to N_0 — a premise whose support came not from the measure itself, but from the aggregate agreement among different determinations.

The four primary quantities with stable measures by 1912 (SS, pp. 287–288):

1. The mean translational kinetic energy W of granules in Brownian motion in dilute emulsions at a given temperature (and, slightly more provisionally, the mean translational kinetic energy per degree Kelvin, W_T). *Perrin, with convergent measures from vertical gradients, translational displacements, and diffusion coefficients.*

2. The value of the fundamental discrete unit of charge e, whether negative or positive, in ions and in such discharges as cathode rays, the photoelectric effect, and α- and β-radioactivity. *Convergent measures by 1912, principally Millikan's oil-drop experiment and the cumulative-charge measure of Rutherford–Geiger (independently confirmed by Regener).*

3. The volume occupied at standard conditions by the electrically neutral helium gas formed by the number of α-particles released and then collected over a given period of time. *Rutherford–Boltwood (1911), building on Rutherford–Geiger's (1908) scintillation-counter results.*

4. The mean energy per degree Kelvin k per discrete absorber-emitter of radiation in the thermodynamically equilibrium state of blackbody radiation, and with it the value of Planck's constant h. *Planck (1900); stable but not yet convergent in 1912, though the Einstein–Debye equation for specific heats of solids offered a route to convergence.*

Note: "While the values determined for each of these 'primary' quantities were theory-mediated in one way or another, *in no case did the theory-mediation include, constitutively, any assumption from molecular-kinetic theory*, and hence any assumption requiring the reality of molecules" (SS, p. 288).

The four corresponding "weak-link" premises (SS, p. 288):

1. The mean translational kinetic energy of granules in Brownian motion at a given temperature equals the mean translational kinetic energy of the molecules in the substrate in which the motion is occurring.

2. The number of unit charges required to produce a mole of a monovalent gas in electrolysis is equal to the number of molecules in a mole.

3. The number of α-particles collected in a given time to form a volume of helium gas is equal to the number of monatomic molecules of helium forming that gas, which in turn is equal to the number of molecules forming any other gas in the same volume at the same temperature and pressure.

4. The mean energy per degree Kelvin per discrete absorber-emitter in blackbody radiation is equal to the mean energy per degree Kelvin of the molecules forming an ideal gas.

These premises are "weak" in two senses:

- *Evidentially weak*: there was no evidence tied specifically to any one determination to support the further assumption in question. Taken in isolation, each premise presupposed — and hence on its face *begged the question of* — the reality of molecules.

- *Informationally weak*: they presupposed "as little about the realm of molecules as they could and still yield a value for N_0" (SS, p. 294). *No* assumption that molecules are rigid spheres, have a definite radius, or possess any particular shape. This is the great virtue of these determinations: they leave many questions about the microphysical realm open, to be answered by subsequent experimental research.

Notice the distinct character of each linking premise:

- The premise for (1) is the most empirically substantive: it presupposes that the *macrophysical* granules have the same mean translational kinetic energy as *microphysical* molecules. (Smith and Seth suggest inaccuracy in this assumption may be the source of the $\sim 10\%$ discrepancy between Perrin's N_0 values and the others.)

- The premise for (2) crosses an electrochemical gap: it equates a count of elementary charges (not in themselves molecules) with a count of molecules per mole, presupposing a one-to-one relation between charge units and molecules of a monovalent gas. Why one unit charge $\rightarrow$ one H_2 molecule rather than one atom or one unit weight of hydrogen?
- The premise for (3) is the *least* challengeable (SS, p. 328): it requires only that α-particles remain discrete and intact as they lose their positive charge to become helium atoms, plus Avogadro's hypothesis relating helium to other gases at standard conditions. (Still: what precisely is it for an α-particle to "lose its charge"? A gap remains between counting α-particles and having a count of helium molecules.)
- The premise for (4) links a thermodynamic property of radiation to a thermodynamic property of gaseous molecules — requiring the equality of mean energies per degree Kelvin across two otherwise unrelated sectors of physics.

A second shared feature: each primary-quantity measure involved reference to something *discrete* and hence countable — granules, unit charges, α-particles, resonators — though the strength of the evidence for discreteness varied (granules are visibly discrete; Rutherford–Geiger actually counted 1–5 "jerks" per minute; Millikan's droplets carried visibly discrete charges; Planck's resonators had the *least* independent evidence as of 1912 and were most controversial). The value for N_0 derived from each primary quantity amounts to a count of discrete items, so the reality of discrete counting units at each stage matters. The *primary count* in each of the four cases is not a count of molecules, but of something else (granules, unit charges, α-particles, resonators); what the agreement among the four shows is that these different counts nevertheless converge on a single definite quantity, namely the number of "molecules" in a mole.

The Aggregate Evidence

Perrin's appeal to the "remarkable agreement" among values of N_0 has been read (e.g., by Salmon and Nye) as common-origin reasoning for molecular reality. Smith and Seth accept the importance of the agreement, but reconstruct its evidential structure more carefully:

- *How much agreement is sufficient*? Philosophical accounts often accept uncritically Perrin's claims of "remarkable agreement." In fact there was considerable spread among the 1908–1913 values, and Perrin's own Brownian motion determination was challenged starting in 1911 and eventually rejected as not in agreement with the others (roughly 10% too high, for reasons still unclear). Moreover, Perrin's 1908 note did not treat the agreement as a *final* argument but as a preliminary comparison, with his determination advertised as "most direct and amenable to unlimited precision."
- *What does agreement among N_0 values provide evidence for*? Smith and Seth identify three distinct contributions (SS, pp. 292–293):
    1. Evidence for the "weak-link" premises licensing each inference from its measured primary quantity to a value for N_0. These premises receive essentially no support from the stability or convergence of the individual measures; their support comes from the aggregate.
    2. Evidence that the diverse determinations, each with its own weak-link premise, were nevertheless measuring — that is, yielding counts of — a *single* quantity.
    3. Evidence showing that a host of separate quantities, some presumed to be constants and others variable, are systematically related to one another not just in theory but in measurment outcomes, thereby providing empirical grounding for molecular-kinetic theory.

- *How the four linking premises support one another*: each presupposes molecular reality in its own way, yet they are *distinct* presuppositions about distinct subject matters (Brownian motion, electrolysis, radioactivity, blackbody radiation). That the four independently derived values of N_0 agree (to within expected error) is evidence that each premise is correctly identifying something real to count, and that "molecules" are the same thing in each case. A good way to see this: what would *failure* of agreement have been evidence against? Not the measures of the primary quantities — those stand on their own — but *the linking premise*. So agreement correspondingly supports the linking premises.

- *The case against van Fraassen's "whole theory" reading*: van Fraassen is correct that theory-mediated measurement requires theoretically posited connections and that uniqueness of coordination must be achieved. But his account treats the whole of molecular-kinetic theory as being tested (or grounded) together. Smith and Seth insist that the actual grounding is far more local: only the *linking premises* receive evidential support from the aggregate of agreeing measurements, and those premises commit to remarkably little about molecular structure.

Virtues of minimal connections to classical kinetic theory:

- *Idealization*: the more substantive models of classical kinetic theory (molecules as rigid spheres, specific shapes or sizes) impose limits on precision, since there is no clear way to remove these idealizations to develop more realistic models.

- *Stability under theoretical change*: the minimal results survived the transition from classical to quantum treatments of atoms and molecules (see SS pp. 303–304). The post-Solvay period saw more and more research built on values of the molecular magnitudes, particularly "generic" quantities that apply to all molecules (N_0, α, e), while attempts to measure shapes of molecules or to give accounts of viscosity, diffusion, thermal conductivity, and deviations from the ideal-gas law remained recalcitrant.

On the Standing of Hypotheses (SS §7.5)

- *Explanatory vs. constitutive roles of hypotheses*:

  Contrast between *explanatory role* (heuristic contribution, intelligibility of phenomena) and *constitutive role* (reasoning that has a conditional character, relying on the hypothesis to make further claims or pose further questions). Determining a hypothesis's constitutive role is often quite subtle.

  Illustrations: (i) explanatory role of the ether, lacking specific details to support further research; (ii) vortex theory of planetary motion in the 17th century vs. Newtonian gravity; (iii) molecular theory before 1900: constitutive role for existence of molecules with *specific* properties (e.g., "radius"). Merely playing a constitutive role does not guarantee success — the latter hypothesis did not support a fruitful line of further inquiry (cf. Meyer 1899, *The Kinetic Theory of Gases*).

- *Taking to be True*:

  > *Rule 4*: In experimental philosophy, propositions gathered from phenomena by induction should be taken to be true, either exactly or very nearly, notwithstanding any contrary hypotheses, until yet other phenomena make such propositions either more exact or liable to exceptions.
  >
  > This rule should be followed so that arguments based on induction may not be nullified by hypotheses.
  >
  > (*Principia*, p. 796)

- Philosophers have often failed to acknowledge an intermediate status for hypotheses prevalent in scientific practice: accepting hypotheses as a basis for further research. Further contrast between *initial* acceptance and *entrenchment*, based on evidence that accumulates as research progresses.
- Current case study: three claims that (as of 1913, SS p. 346) should be taken to be true — (i) all moles of any substance contain $N_0 \approx 6.4 \times 10^{23}$ molecules; (ii) molecules in a gas have the same mean translational kinetic energy per degree Kelvin (around 1.95×10^{-16} ergs); (iii) there is a definite fundamental, universal unit of negative electric charge of about 4.5×10^{-10} esu, combined with a definite fundamental mass. Prior to 1908, no comparably persuasive argument for taking the "radius of molecules" or other features of molecules to be true.
- *Initial* acceptance is based on relatively sparse evidence. Pragmatic aspect of the rule: advantages to "taking to be true" as basis for further research, *despite* initially weak evidence.
- *Failure modes*: (i) the proposition does not hold at expected level of precision, or no clear way to evaluate precision; (ii) the proposition holds only for a restricted domain; (iii) the proposition is a "surrogate" for a proposition stated in completely different terms. (P' as surrogate for P: many inferences employing P remain valid when P' is inserted in its place. Intended as an alternative to seeing P as an "approximation" to P'.)
- *Gains from taking propositions to be true*: turn available data into richer evidence, e.g. by establishing links among disparate phenomena; enable experiments and observations to answer open questions. How to mitigate the risks? (See SS pp. 356–357.)

- *Contrast with Inference to the Best Explanation*:

  Smith and Seth seek to distinguish "taking to be true" from IBE. Two sides: first, contrast with IBE — to show what is involved beyond IBE; second, a positive account of the stronger status. Their positive account contrasts the kind of case that could have been made for molecules before 1908 (as by Meyer) — where all proposed explanations *had* to make substantive assumptions (e.g., spherical molecules) — with the consensus from around 1913. Considerations:

  1. *Specificity*: how can the proposition be leveraged into further results? Can it be built on?
  2. *More than mathematical tractability*: does the hypothesis beg important questions, or does it instead set the stage for experimental answers? (Contrast is clear: the assumptions needed for N_0, e, α do not prejudge important issues.)
  3. *Likelihood of survival under theoretical change* (three conditions, SS p. 356): meant to ensure that research assuming the claims will not be discarded.
  4. What this justifies is a form of "realism" about molecules without claiming that everything about them has been settled.